

Troubleshooting Guide: Distillation Column — Overhead Product Off-Spec

Category: Separation Equipment / Distillation **Unit:** Distillation Column (tray or packed, electrically heated reboiler) **Tools:** UniSim rigorous column model with reboiler duty sensitivity analysis **Fluid Package:** Peng-Robinson (PR) or SRK, depending on the hydrocarbon system (specific package not stated in original case notes) **Symptom:** Overhead product gradually drifting off spec despite stable feed and normal-looking steam pressure

1. Quick Reference

Item	Detail
Reported symptom	Overhead product gradually drifted off specification
Initially unclear	Whether this was a process deficiency (insufficient reboiler duty) or an instrumentation/equipment fault hiding behind a normal-looking reading
Actual root cause	Malfunctioning thyristor (SCR power controller) delivering only ~70% of required power to the electric reboiler heater — while the operator display showed the correct setpoint
Fix	Replace the thyristor
Diagnostic signal	Simulated overhead composition (using actual feed data + reboiler duty sensitivity) matched plant data only when duty was reduced — pointing at duty, not feed or trays
Prevention	Periodic checks that verify actual heater duty against controller output, not just the setpoint display

2. Symptom

- Overhead product **gradually drifted off specification** — not a sudden step change, which matters diagnostically (see Section 3).
- **Feed conditions were stable.**
- **Steam pressure looked normal** — except this unit uses an **electrically heated reboiler**, so “steam pressure” here refers to a related utility/indicator that was not, in the end, the limiting factor (see Root Cause).

3. Why This Wasn’t a Simple Feed or Tray Problem

A **gradual** drift with **stable feed** and a **normal-looking heat source indicator** creates ambiguity between two very different fix paths:

1. **Process deficiency** — insufficient reboiler duty, genuinely not enough heat being delivered to generate the vapor traffic needed for separation.

2. **Instrumentation/equipment problem** — the heat source indicator is telling the operator everything is fine, while the actual delivered energy is not what the display shows.

These require completely different troubleshooting paths (steam/heater-side vs. electrical/controller-side), so before dispatching a field team, the priority was determining **which one** was actually happening — using simulation to test the “insufficient duty” hypothesis quantitatively rather than guessing.

4. Diagnostic Approach

Step 1 — Build a rigorous column model with actual operating data

The column was modeled in **UniSim**, using: - Actual feed composition - Actual operating data (pressures, temperatures, flows)

Fluid package note: PR and SRK are both standard choices for hydrocarbon distillation VLE. PR is generally preferred for systems with **lighter components or higher pressure**; SRK tends to be preferred for **closer boiling-point separations at moderate pressure**. The specific package used in this case wasn’t recorded in the original notes — when reproducing this type of analysis, choose based on your system’s component slate and operating pressure rather than defaulting to one package by habit.

Step 2 — Run reboiler duty sensitivity analysis

Rather than assuming the reboiler duty shown/implied by the steam-side reading was correct, a **sensitivity study on reboiler duty** was run against overhead composition.

Hypothesis tested: Is reduced vapor generation (lower effective reboiler duty) sufficient to explain the observed loss of overhead purity?

Step 3 — Compare simulated composition to plant measurements

The simulation showed that **reduced vapor generation from lower reboiler duty directly explained the loss of overhead purity**, and the predicted composition **matched plant measurements closely** at a reduced-duty case.

Interpretation: This is the same core diagnostic pattern used in the TEG dehydration and reciprocating compressor cases — build a model on actual data, then see what condition makes the model agree with reality. Here, the model only matched plant behavior when duty was reduced below its nominal/expected value — which meant duty (not feed composition, not tray/packing performance) was confirmed as the actual driver.

Step 4 — Redirect the investigation based on the model’s conclusion

Because the model confirmed duty was insufficient — not feed or separation efficiency — the field team was directed toward an **electrical inspection of the reboiler**, rather than continuing to troubleshoot the steam side, which had already looked normal and would likely have been a dead end.

Quantitative Basis

- Overhead purity spec: ≥ 95 mol%. Baseline held near 96 mol%, drifted to **89 mol% over 3 weeks**.

- Reboiler design duty: 4.2 MMBtu/hr. UniSim sensitivity analysis found that matching the observed 89 mol% overhead composition required a reboiler duty of **2.9-3.0 MMBtu/hr — roughly 70% of design.**
- Following the sensitivity result, direct electrical measurement at the heater element found current of **145 A, against a rated/commanded 205 A (71% of rated)** — despite the controller display showing 100% commanded output. This closely matched the ~70% duty shortfall the model had already identified.

5. Root Cause

A **malfunctioning thyristor (SCR power controller)** was delivering only ~70% of **required power** to the electric heater. Critically, **the operator display showed the correct setpoint** — the instrumentation the operators were trusting was not reflecting the actual power being delivered. This is a case of the *commanded* value and the *actual delivered* value diverging without any alarm or visible discrepancy at the operator interface.

6. Corrective Action

1. Replaced the faulty thyristor.
2. Verified restored reboiler duty.
3. Confirmed overhead product purity returned to specification.

7. Verification

- Post-replacement, heater current measured **203 A — 99% of the 205 A rated value.**
- Reboiler duty restored to **4.15 MMBtu/hr**, close to the 4.2 MMBtu/hr design value.
- Overhead purity recovered to **96.2 mol%**, within spec, and held there over the following **30 days.**

8. Prevention / Long-Term Fix

- Recommended **periodic checks that verify actual heater duty against controller output**, rather than relying on the setpoint display alone.
 - This closes the specific gap that caused the extended drift: a setpoint display that looked correct while the actual delivered energy silently degraded.
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9. General Troubleshooting Checklist (Reusable)

Use this checklist for future distillation column gradual off-spec product events:

- ☐ Confirm feed composition and flow are stable (rules in/out a feed-driven cause)
- ☐ Note whether the drift is **gradual** (favors equipment degradation) or a **step change** (favors an operational/setpoint change)
- ☐ Do not treat a normal-looking utility-side indicator (steam pressure, setpoint display) as proof that duty is actually being delivered — indicators can be correct on the command side and wrong on the delivery side
- ☐ Build/update a rigorous column model using actual feed composition and operating data
- ☐ Run a **reboiler duty sensitivity analysis** against the off-spec product composition
- ☐ If a reduced-duty case matches plant data closely, treat **duty delivery** as confirmed — not feed, not tray/packing efficiency
- ☐ For electrically heated reboilers, direct inspection toward the **power controller/SCR/thyristor**, not just the heater element itself

- ☐ For steam-heated reboilers, the equivalent check is control valve position vs. actual steam flow, not just steam header pressure
- ☐ After repair, verify fix by confirming **actual duty delivered**, not just that the setpoint display reads correctly
- ☐ Add a standing check that periodically cross-verifies actual delivered duty against commanded/setpoint duty

10. Key Takeaway

A setpoint display shows what the system was **told** to do, not necessarily what it's **actually doing**. When a gradual off-spec trend shows up despite everything upstream looking stable, use a duty (or equivalent) sensitivity analysis to test whether reduced delivered energy — not feed, not separation efficiency — explains the data. If it does, inspect the delivery path (controller, thyristor, valve actuator) directly, because the indicator that looks “normal” may only be reporting the command, not the result.

Related Concepts / Tags

distillation-column overhead-product off-spec reboiler-duty sensitivity-analysis thyristor SCR-controller electric-reboiler UniSim Peng-Robinson SRK instrumentation-fault setpoint-vs-actual

This guide is derived from a real field troubleshooting case. Values and thresholds shown are specific to the reported case and should be validated against your own unit's design basis before applying. Note: the specific fluid package used in the original case was not documented; PR and SRK are cited as the standard candidates for this service.